

Sample exam

Task 1 (4 points)

Compute the Laplace transform of $f(t) = e^{(-2t)}$.

Task 2 (6 points)

The figure shows a finite automaton over the alphabet $\{0, 1\}$.

[figure]

- a) Which words of length two does it accept?
- b) Draw an automaton for the reversed language.

Task 3 (5 points)

Sketch the Bode plot of $G(s) = 10 / (s + 1)$.

Solutions

Task 1: $F(s) = 1 / (s + 2)$

Task 2a: only 01.

Task 2b: C -1-> B -0-> A, with C the start state and A accepting.

Task 3: A horizontal line at 20 dB up to $w = 1$, then -20 dB per decade.